International Isotope Society UK Group

32nd Annual Symposium

The Synthesis and Applications of Isotopically Labelled Compounds

Friday, 13th November 2026
The Møller Institute, Cambridge

https://www.iis-uk.org

Sponsors

The organising committee would like to thank the following companies for their generous sponsorship:

Welcome & Acknowledgements

On behalf of the organising committee, it is my pleasure to welcome you to the 32nd annual symposium of the International Isotope Society UK Group on the synthesis and applications of isotopically labelled compounds held, once again, at The Møller Institute in Cambridge.

Professor Gill Reid from the University of Southampton will open the meeting with a talk titled "*Exploring Metal-Macrocyclic Scaffolds for Binding Fluorine-18.*" Our afternoon session will be opened by Professor Troels Skrydstrup from the University of Aarhus in Denmark with a talk titled *"Carbon Isotope Labeling Techniques Involving Organometallic Intermediates.*"

Today's programme will also cover a multitude of topics including ^{68}Ga radiopharmaceuticals, carbon chain homologation, ^{14}C-enabled clinical studies with AMS, radiochemical stability testing, catalysis and CO_2 transfer reactions. I hope you'll find plenty of topics of interest.

I would like to thank all our speakers and those presenting posters for joining us and taking the time to share their work. I'd also like to thank our sponsors and exhibitors for their generous support both this year and in years past. My thanks also go to our session chairs and to the staff of The Møller Institute for helping us to stage this event. This meeting would not have been possible without the dedication and enthusiasm of the organising committee — my thanks go to all involved.

Lastly, I have to thank you all for attending and helping to make this event so successful. I hope you have a very enjoyable day.

Chris Winfield.
Chair, IIS UK Group.

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Scientific Programme

8:30 am Registration, morning coffee & manufacturers' exhibition.

9:00 am Welcome: Dr Chris Winfield (Eurofins Selcia, Chair of the IIS UK Group).

Morning Session 1

Chair: Dr Graham Smith (University of Cambridge, UK).

9:05 am **Professor Gill Reid** (University of Southampton, UK).
"*Exploring Metal-Macrocyclic Scaffolds for Binding Fluorine-18.*"

9:55 am **Dr Graeme McRobbie** (Bicycle Therapeutics, UK).
"*Gallium-68 Radiopharmaceuticals: An Industry Perspective.*"

10:20 am *5-Minute Flash Presentations*

Isabella Ogilvy (University of Oxford, UK).
"^{18}F-Radiopharmaceutical Diversification Enabled by Deaminative Cross-Electrophile Couplings."

Matthew Hird (University of Cambridge, UK).
"Production of [^{11}C]MRB on Synthra MeI plus in a GMP environment."

Maddison Lovell (University of Oxford, UK).
"A Ni-Mediated Cross-Coupling Approach to Deuterated ^{18}F-Fluoromethylated (Hetero)arenes."

Ridhima Raina (GSK & University of Strathclyde, UK).
"Photochemical Approaches Towards Late-Stage Methylation for Isotope Labelling."

10:50 am Manufacturers' exhibition, poster session, tea & coffee.

Morning Session 2

Chair: Dr Daniela Roman (GSK, UK).

11:20 am **Dr Mattia Silvi** (University of Nottingham, UK).
"*Homologating Carbon Chains to Unlock New Reactivity*."

11:45 am **Dr Marie Croft** (Pharmaron, UK).
"*Application of Accelerator Mass Spectrometry in Support of ^{14}C-Enabled Clinical Studies.*"

12:10 pm Buffet lunch, manufacturers' exhibition & poster session.

Scientific Programme

Afternoon Session 1

Chair: Dr David Lindsay (University of Strathclyde, UK).

1:25 pm **Professor Troels Skrydstrup** (University of Aarhus, Denmark).
"*Carbon Isotope Labeling Techniques Involving Organometallic Intermediates*."

2:15 pm **Dr Antonio Moreno-Herrera** (AstraZeneca, UK).
"*Suitability of the Extrapolation of Radio-labelled Drug Substance Stability Data for the Determination of Extended Retest Periods.*"

2:40 pm Manufacturers' exhibition, poster session, tea & coffee.

Afternoon Session 2

Chair: Dr Helen Betts (Nottingham University Hospitals NHS Trust, UK).

3:15 pm **Dr Allegra Franchino** (University of Durham, UK).
"*Repurposing H-bond Donors for Transition-Metal Catalysis.*"

3:40 pm **Dr Gregory Perry** (University of Southampton, UK).
"*CO_2 Transfer Reactions for Carboxylation and Carbon Isotope Labelling.*"

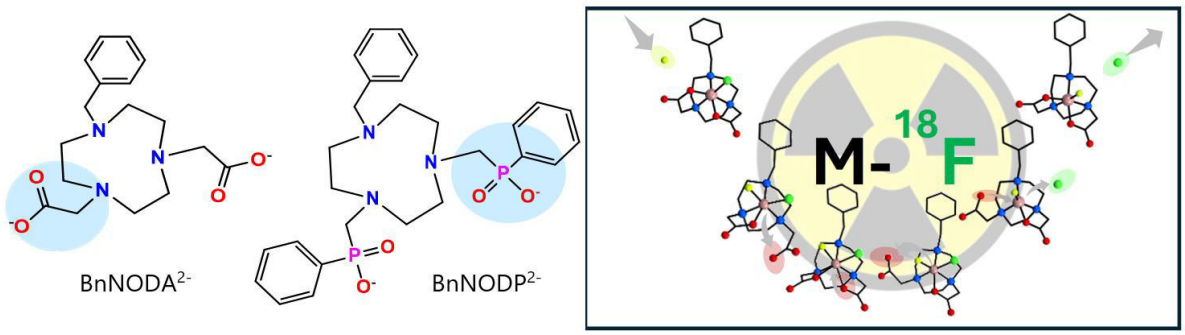
4:05 pm Concluding Remarks: Dr Chris Winfield (Eurofins Selcia, Chair of the IIS UK Group).

Presentation Abstracts

Exploring Metal-Macrocyclic Scaffolds for Binding Fluorine-18

Professor Gill Reid.
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Coordination complexes containing fluoride co-ligands often display markedly different properties from their heavier halide analogues. Over the past 15 years, this distinction has driven substantial interest in medium- and high-oxidation-state *d*- and *p*-block fluoride complexes.[1–3]
A major impetus for much of the recent work has been the prospect of combining fluorophilic metal ions with the rapid association and slow dissociation kinetics characteristic of macrocyclic ligands. While metal–fluoride complexes incorporating fluorine-18 (^{18}F; $t_{½}$ = 110 min; β^+) are advancing PET imaging, and creating new opportunities for Total Body PET and theranostics, the factors governing metal–chelator selection, radiofluorination mechanisms, and hydroxide competition remain unclear. Although several Al–^{18}F tracers based on NOTA and NODA chelators have reached clinical use,[4,5] mechanistic understanding is limited and has hindered extension to other metals and chelators, as well as understanding of the relative merits of "one-pot" (radio)fluorination protocols *vs*. late-stage (radio)fluorination of pre-formed complexes.
This talk will present our recent studies of trivalent metal complexes supported by tacn-based N_3-donor macrocycles bearing a range of pendant-arm functionalities, e.g. Bn-NODA and Bn-NODP. These highly tunable scaffolds are being explored for binding radiofluorine and with the aim of enabling late-stage radiofluorination under mild conditions.



By integrating experimental studies with density functional theory (DFT) calculations,[6-9] we are examining how the identity of the metal ion and the nature of the pendant arms govern both the formation and stability of the corresponding metal–fluoride complexes. These complementary approaches provide valuable mechanistic insight into the factors controlling fluoride binding and help inform the design of robust, readily fluorinated scaffolds for future ^{18}F-radiotracer development.

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Development of Gallium-68 Radiopharmaceuticals: An Industry Perspective

Dr Graeme McRobbie.
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The expanding landscape of precision oncology demands novel imaging agents capable of providing rapid, high-contrast visualisation of clinically relevant targets. While antibodies and small molecules have limitations in pharmacokinetics or specificity, Bicycle® peptides represent a unique, clinically validated modality. Their small size allows for rapid tissue penetration and fast renal clearance, while their large binding surface area enables antibody-like affinity and specificity. Bicycle Radionuclide Conjugates (BRCs) are designed to utilise Bicycle targeting vectors to deliver radioisotopes for both diagnostic imaging and targeted radiotherapy (theranostics).

Utilising a proprietary phage display platform, potent binders are identified and lead candidates are optimised for affinity, stability, and pharmacokinetic properties. Optimised peptides are then functionalised with chelators for site-specific radiolabeling with isotopes such as gallium-68 (Figure 1). These BRCs are then evaluated *in vitro* for binding affinity and specificity, and *in vivo* via PET imaging as well as ex vivo biodistribution studies in relevant tumor xenograft models.

Once the optimised candidate has been identified, the drug product development process begins with the aim of providing a robust manufacturing process for a BRC® drug product that meets the established specifications.

The Bicycle® platform has proven to be a robust, scalable, and highly effective engine for the discovery and development of novel radiopharmaceuticals. The unique properties of bicyclic peptides translate directly into superior agents with rapid, high-contrast tumor visualisation and favorable tumor to background profiles. This has the potential to deliver a pipeline of best-in-class PET diagnostics and radiotherapeutics across a range of potentially high value cancer targets to address critical unmet needs in patient stratification and treatment monitoring across a breadth of cancer indications.

This presentation will summarise some preclinical aspects and then describe the development of a robust manufacturing process for a gallium-68 BRC® drug product that has been established to enable reliable supply of drug product for early phase clinical studies.

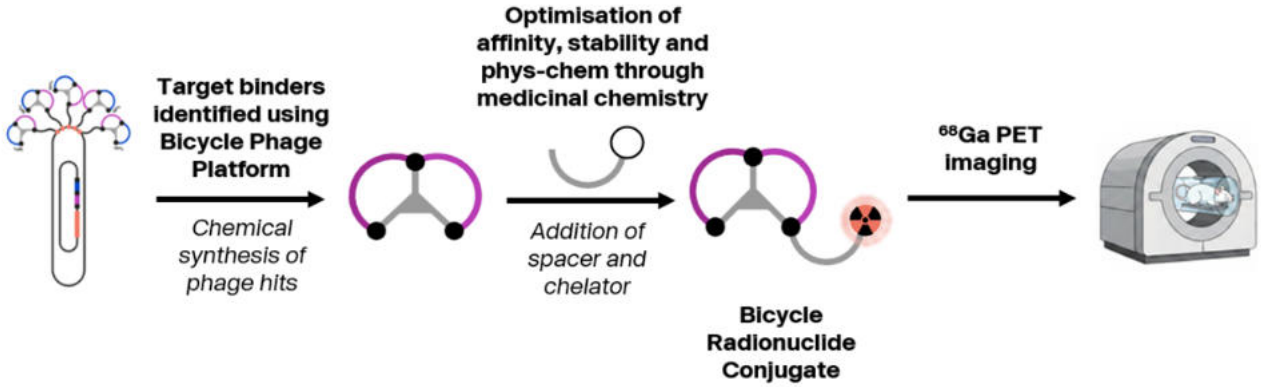


Figure 1. Overview of the identification and design of Bicycle® Radionuclide Conjugates (BRCs).

Homologating Carbon Chains to Unlock New Reactivity

Dr Mattia Silvi.
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Methylene homologues play fundamental roles in chemistry and biology.[1] Thus, the homologation reaction—i.e., the elongation of carbon chains by a single carbon unit—has been the subject of extensive study, establishing it as a fundamental process in synthetic chemistry. However, classical methods often involve multistep routes and harsh conditions, limiting the scope of application of this important transformation.[2] Inspired by our previous research about highly polarized reagents in radical chemistry,[3] my research group has recently introduced methods to address long-standing challenges in homologation chemistry, and extend its applications beyond conventional uses. In this presentation, I will describe a direct and iterative one-carbon homologation protocol for unmodified carboxylic acids, a transformation that had previously remained elusive in synthesis.[4] I will further demonstrate how this approach can constitute a versatile tool for the user-defined extension of carbon chains in complex bioactive molecules.
Finally, I will show how the concept of homologation can be extended in the context of functional-group activation chemistry, to access unconventional patterns of substitution. This strategy opens a route to 1,3-disubstituted products in alkene difunctionalization chemistry,[5] and enables generalized access to pharmaceutically relevant 2-substituted four-membered carbo- and heterocycles from olefins.[6]

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Application of Accelerator Mass Spectrometry in Support of ^{14}C-Enabled Clinical Studies

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Accelerator Mass Spectrometry (AMS) has been increasingly used in drug development for the detection of low levels of ^{14}C-labelled compounds. While historically utilised for detection of ^{14}C-labelled compounds with long half-lives, low exposure, or in microdosing studies, its use has expanded following industry recommendations to incorporate ^{14}C microtracers into Phase 1 human absorption, metabolism and excretion (AME) and human absolute bioavailability (ABA) studies,[1] together with FDA mass balance guidance.[2]
Early AMS systems were large, with estimated footprints of 250 m^2, required specialist operators and relied on labour-intensive sample preparation processes including conversion of the sample to elemental carbon (graphitization). Advances in instrumentation have led to smaller, more user-friendly systems with the introduction of automated gaseous (CO_2) interfaces, significantly improving usability and efficiency, such as the Ionplus Low Energy AMS (LEA), which offers simpler sample processing and faster throughput.
This presentation will provide an overview of the recent advances in AMS technology as well as its role in supporting ^{14}C-enabled clinical studies. The evolution of the technology will be explored, from the very large instruments requiring time intensive graphite preparation methods, to the smaller directly interfaced CO_2 instruments, as well as the applications of the technology in support of ADME and ABA studies.

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Recent Carbon Isotope Labelling Techniques Involving Organometallic Intermediates

Professor Troels Skrydstrup.
Villum Investigator. Carbon Dioxide Activation Center, Novo Nordisk Foundation CO2 Research Center, Department of Chemistry, Aarhus University, Denmark.
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In this talk, I provide a short overview on the successful collaboration between the isotope chemistry team at AstraZeneca and my research group over the last 15 years dealing with late-stage carbon isotope labelling of bioactive molecules. I will talk about the use of carbon monoxide surrogates for performing safe and stoichiometric carbonylation chemistry applying transition metal chemistry with the involvement of organometallic intermediates. I will also discuss efforts to develop a molecular surgery strategy comprising of a sequence of C–C bond cleaving and bond forming events for the insertion of a carbon label, as well as our recent collaboration between both teams on the adaption of the carbonylation chemistry to the selective late-stage carbon-isotope labelling of peptides.

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Evaluation of the Suitability of Extrapolating Radio-labelled Drug Substance Stability Data for the Determination of Extended Retest Periods

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Associate Principal Scientist. Early Chemical Development, Pharmaceutical Sciences, R&D, AstraZeneca, Macclesfield, UK.
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Learning objective:

Stability extrapolation of radio-labelled drug substance has been demonstrated to be suitable for the determination of extended retest periods and provide insights into stability risks for clinical trial support

Abstract:

Radio-labelled drug substance stability studies are essential for supporting human Absorption-Distribution-Metabolism-Excretion (hADME) studies. For this purpose, a stability study of the investigated radio-labelled compound is conducted, and a retest period is assigned based on real stability data derived from the most recent timepoint. As new stability timepoints become available, the retest period is updated accordingly. This approach deviates from the typical process followed for non-labelled compounds, which employs stability extrapolations and extended retest periods based on principles outlined in ICH Q1E Evaluation for Stability Data guidelines.

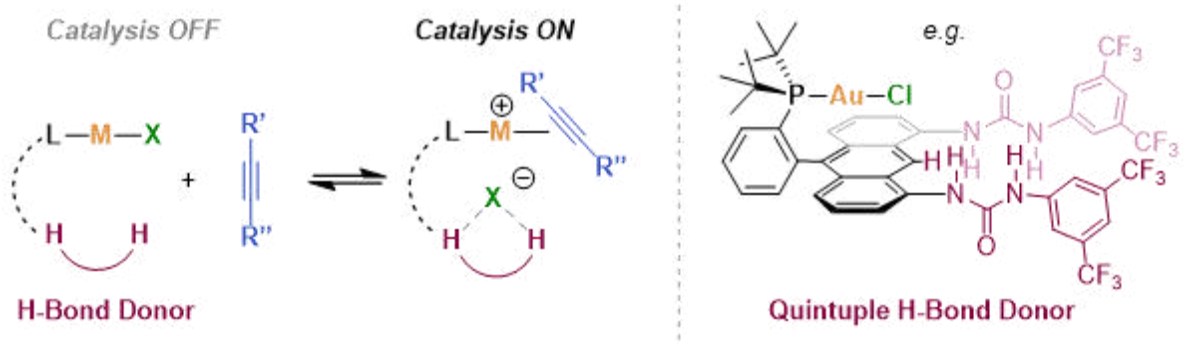
This presentation assesses the suitability of stability data extrapolations for C-14 labelled compounds following recommendations described in ICH Q1E guidelines. We analyzed stability data from six C-14 labelled compounds tested at AstraZeneca over the past 10 years at specific activities ranging 1-81 µCi/mg. The data were extrapolated using predictive models to determine extended retest periods, which were later confirmed against actual experimental results. In all studies, the extended retest periods were fully consistent with the observed results.

We propose that stability extrapolations could be utilized to establish extended retest periods for radio-labelled compounds. This will significantly simplify the current release process for radio-labelled compounds which is cyclically repeated after any retest period update.

Repurposing H-Bond Donors for Transition-Metal Catalysis

Dr Allegra Franchino.
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In this talk, I will describe our efforts to pioneer conceptually new strategies that combine transition-metal catalysis (chloride scavenging for precatalyst activation) and supramolecular chemistry (H-bond donors for anion binding).
Gold catalysis provides access to a remarkable array of complex carbo- and heterocyclic products from simple starting materials. However, the use of silver salts to activate gold(I) chloride precatalysts can be problematic, due to Ag(I) light sensitivity, hygroscopicity, redox activity and interference with the desired gold chemistry, which would otherwise be water- and air-tolerant. Self-activating gold(I) chloride complexes have emerged as a promising alternative, but still suffer from much lower activity and narrower applicability than silver salts, as Au–Cl cleavage is rate limiting.[1,2]
Addressing these limitations, we have recently reported a self-activating Au(I) chloride complex with a modified Buchwald ligand that incorporates a supramolecular chloride receptor: a 1,8-anthracene bisurea quintuple H-bond donor. This ligand design enables for the first time H-bond donors to compete with inorganic chloride scavengers in terms of activity and generality.[3]
More recent results from our group focus on multidentate H-bond donors that are no longer tethered to the ligand, can activate various metal–pseudohalide bonds and are applicable to enantioselective reactions.[4]

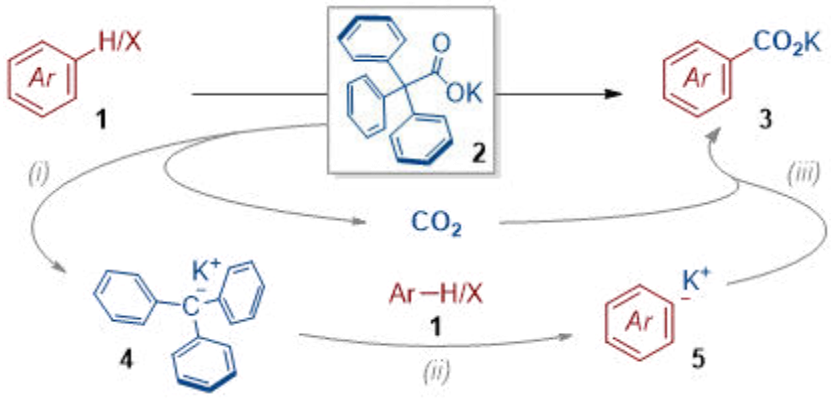


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CO_2 Transfer Reactions for Carboxylation and Carbon Isotope Labelling

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CO_2 transfer reactions have been known since at least the 1950s when Raecke reported the disproportionation of aromatic carboxylates.[1] However, the requirement for elevated temperatures (>350 °C) has limited their applicability. This talk will summarise recent developments in CO_2 transfer, which has led to more general carboxylation reactions and applications in carbon isotope labelling.[2] The talk will feature potassium triphenylacetate **2**, a CO_2 transfer reagent that originated from our laboratory.[3] The general mechanistic aspects of this reactivity includes 1) decarboxylation of carboxylate **2** to give trityl anion **4** and CO_2 in situ (step i); 2) metalation of substrate **1** by the organometallic species **4** via deprotonation or metal-halogen exchange in a process we term decarboxylative metalation (step ii); and 3) the capture of the in situ generated CO_2 by intermediate **5** to deliver the carboxylated product **3** (step iii). The talk will reveal how these reagents offer unique reactivity for organic synthesis and enhance current methods in carboxylation and carbon isotope labelling.

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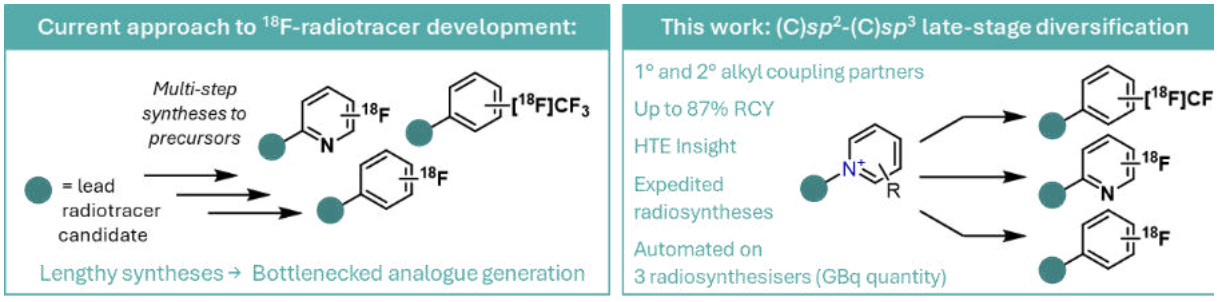
Flash Presentation Abstracts

^{18}F-Radiopharmaceutical Diversification Enabled by Deaminative Cross-Electrophile Couplings

Isabella Ogilvy,[1] Joseph Ford,[1] Sebastiano Ortalli,[1] Evelien Renders,[2] Thomas Hayes,[2] Shuanglong Liu,[2] Inne Mortiers,[2] Anastasia Nikolopoulou,[2] Alexandre Sorlin,[2] Andrés Trabanco,[2] Matthew Tredwell,[3] Peter Buijnsters,[2] Rhys Salter,[2] Véronique Gouverneur[1]*
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Key words: *fluorine-18 radiochemistry, cross-couplings, high-throughput experimentation*

The development of fluorine-18 radiotracers is central to clinical positron emission tomography (PET) imaging and supporting drug discovery campaigns. However, the preparation of complex radiolabelling precursors often limits the rapid generation and evaluation of structural analogues, restricting exploration of radiochemical space. This is highly prevalent in the syntheses of specialized precursors amenable to site-selective, late-stage radio fluorination with weakly nucleophilic [^{18}F]fluoride (^{18}F; $t_{1/2}$ =109.7 mins; 97% β+ decay; 635 keV).[1]
In this work, we present a nickel-mediated aryl (C)sp^2–(C)sp^3 cross-coupling strategy compatible with radiosynthesis, employing readily-accessible amine-derived 2,4,6-triphenylpyridinium salts as alkyl coupling partners.[2] In analogy to pharmaceutical discovery, this approach enables late-stage diversification from common precursors, reducing synthetic burden and facilitating access to structurally diverse ^{18}F-radiotracers. The method is applicable to both primary and secondary *N*-alkyl pyridinium salts, achieving high radiochemical conversions (up to 86%). A tailored non-radiochemical high-throughput experimentation (HTE) assay was developed to rapidly identify optimal Ni-ligands for structurally diverse fluorinated (hetero)aryl iodides, expediting reaction optimisation. Crucially, the assay results translated well to analogous radiochemical systems.
A diversification case study targeting GSK-3 kinase inhibitors demonstrates the utility of this strategy, enabling the synthesis of six ^{18}F-labelled analogues from a single precursor, including production at gigabecquerel (GBq) scale across three commercial synthesis platforms. This work highlights a novel approach for accelerating radiotracer development through modular and scalable late-stage functionalisation.



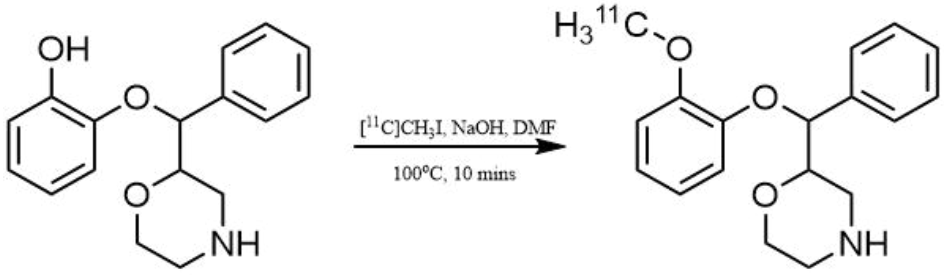
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Production of [^{11}C]MRB on Synthra MeI Plus in a GMP Environment

Matthew Hird[1], João Louçano[1], Sebaey Mahgoub[1], Aniruddha Doke[1], Franklin I. Aigbirhio[2] and Graham Smith[1]
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Introduction: [^{11}C]Methylreboxetine (MRB), is a PET radiopharmaceutical recently validated at the Radiopharmaceutical Unit (RPU). It is used in clinical settings for diagnosis of progressive supranuclear palsy (PSP) which is often misdiagnosed as Parkinsons due the similarity of the clinical symptoms.
Materials and Methods: The Synthra MeIplus is used for the synthesis of [^{11}C]MRB. [^{11}C]CO_2 is produced from the GE PETtrace 800 cyclotron in 50 min. Irradiation produces ~45-50 GBq of [^{11}C]CO2 radioactivity which is converted via [^{11}C]CH_4 to [^{11}C]CH_3I. The radiosynthesis of [^{11}C]MRB proceeds via methylation of the MRB precursor (0.5 mg) in a NaOH/DMF solution. After approx. 1 minute delivery to the reactor, the reaction is heated at 100°C for 10 minutes. The solution is cooled then diluted with water before being injected onto the semi-preparative column. The product peak is collected in water and immobilised on an MCX cartridge before washing with water. Pure [^{11}C]MRB is then eluted from the cartridges with ~1 mL of Ethanol and 1 mL of PBS into a solution of 14 mL 0.9 % NaCl, then delivered to the dispensing system (Clio dispenser). For quality control, a 10-minute HPLC method was employed using isocratic elution with 35% THF in water at a flow rate of 1 mL/min, with detection at 235 nm, on a C18 column to confirm radiochemical identity and purity.



Scheme 1. Radiosynthesis of [^{11}C]MRB

Results: Currently in routine production at the RPU for one patient dose at a time. Typical synthesis time is 60 minutes, and average non-decay radiochemical yield was 11 %, (n = 4, from [^{11}C]CH_3I).
Conclusions: [^{11}C]MRB has been validated for clinical production in RPU. The manufacturing/QC process is robust, semi-automated with minimal user interaction and provides enough radiotracer for 1 patient dose on-site.

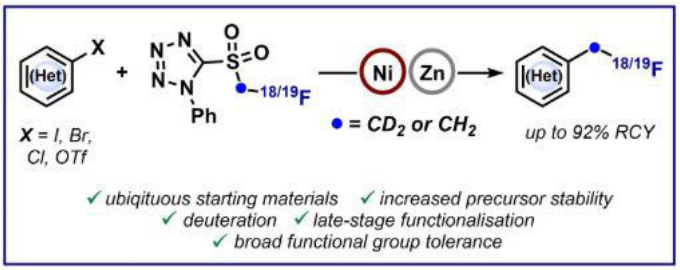
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A Ni-Mediated Cross-Coupling Approach to Deuterated ^{18}F-Fluoromethylated (Hetero)arenes

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The (hetero)aryl-^{18}F-fluoromethyl group ([^{18}F]CH_2F) is highly prevalent in ^{18}F-containing PET radiotracers.[1] However, radiotracers bearing a [^{18}F]CH_2F group can be susceptible to *in vivo* degradation by radiodefluorination.[2] Practitioners have therefore exploited deuteration of the fluoroalkyl chain as a tool for increasing stability whilst maintaining biological activity.[3] To date, radiosynthetic strategies to access deuterated ^{18}F-fluoroalkyl compounds demand the multi-step synthesis of bespoke labelling precursors, bearing preinstalled leaving groups, such as (pseudo)halides, that can be displaced with [^{18}F]fluoride. In addition to synthetic challenges arising from the late-stage introduction of reactive groups, careful planning is required to ensure the isotopic integrity of the deuterated precursor is maintained in further synthetic manipulations. Moreover, precursors bearing reactive leaving groups often also suffer from stability issues.[1] Contrastingly, only a single protocol for the direct installation of $CH_2{}^{18}F$ onto aromatic substrates has been disclosed to date, and this was not extended to deuteration.[4]



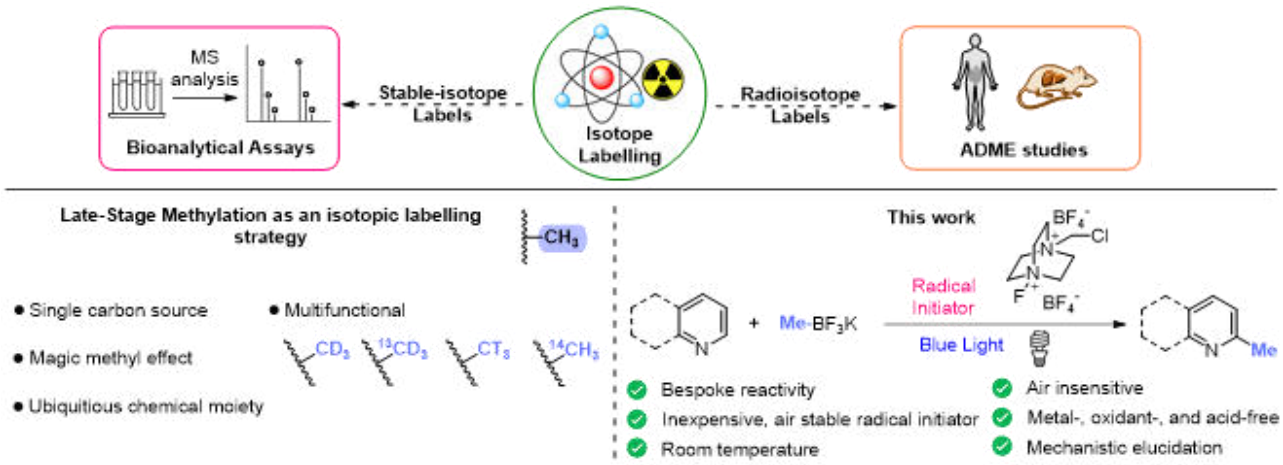
With these challenges in mind, we have developed a novel platform for ^{18}F,D_2-fluoromethylation, *via* a cross-electrophile coupling between aryl (pseudo)halides and a suitable deuterated reagent.[5] This approach offers a series of key advantages over traditional methodologies, such as the use of readily available starting materials, and modular installation of [^{18}F]CD_2F, allowing for rapid library generation for PET ligand discovery. The same ^{19}F-fluoromethyl containing reagent is also competent in the analogous non-radioactive reaction, for the preparation of reference standards for quality control and analysis. This technology has been exemplified on a broad substrate scope and has been extended to (semi)automation for the isolation a ^{18}F,D_2-containing PARP radiotracer.

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Photochemical Approaches Towards Late-Stage Methylation for Isotope Labelling

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Isotopically labelled molecules are highly valued within the pharmaceutical and agrochemical industry, serving as internal standards during lead discovery, and as metabolic markers in preclinical and clinical studies.[1] Currently, however, their synthetic routes are often time-consuming and expensive, due to the limited availability of (radio)labelled precursors and the requirement for radioactive waste management. Late-stage labelling offers a versatile and efficient alternative to conventional multistep synthesis by introducing isotopic handles in the final stages, leading to enhanced efficiency and broader scope. The methylation of *N*-heterocycles represents a particularly advantageous late-stage technique, given the ubiquity of methyl[2] and *N*-heteroarenes[3] in drug molecules, and the potential to incorporate carbon and hydrogen isotopes *via* a unified method, through selection of an appropriately labelled methylating reagent. This project aims to develop a bespoke, catalyst-, oxidant-, and acid-free photoredox-mediated methylation strategy for late-stage labelling using a simple, inexpensive, air-stable radical initiator, and $MeBF_3K$[4] as the methylating agent.

A Design of Experiments (DoE) approach, combined with a high-throughput experimentation (HTE) workflow, yielded the optimised reaction conditions. Extensive mechanistic investigations have identified the photoactive species and elucidated the role of the various reaction components. Ongoing studies have deepened these insights, and the reaction has been used to methylate a range of substrates, including a drug-like scaffold. This strategy, therefore, provides a flexible, safer and cost-effective route to the installation of a range of methyl isotopologues, enabling broad application across academia and industry.

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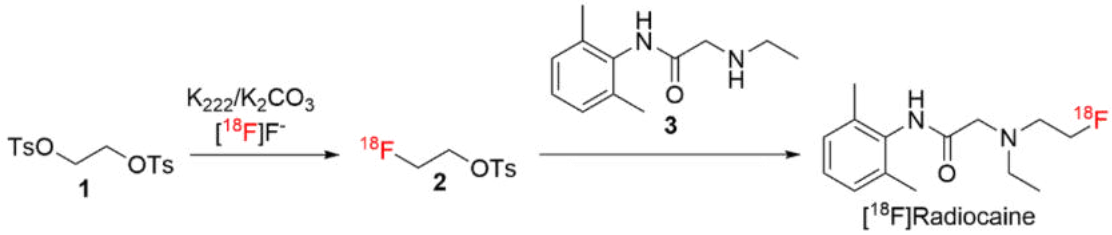
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Poster Abstracts

From Trace Yields to Clinical-Scale Batches: High-Throughput Screening-Guided Development of a Fully Automated, Dual-HPLC Radiosynthesis of [^{18}F]Radiocaine

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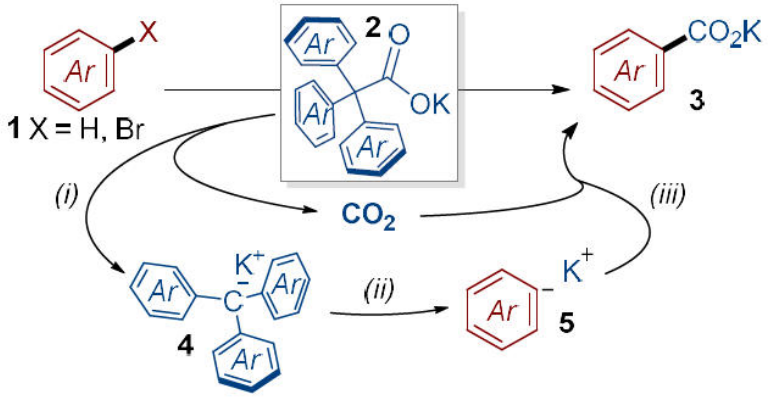
Objectives: Voltage-gated sodium channels (NaVs) underlie action-potential generation in excitable cells, and altered NaV expression or function has been linked to a broad range of pathologies including chronic pain, as well as cardiac and neurological disorders. [^{18}F]Radiocaine is a PET imaging agent targeting NaVs and has the potential to improve the way millions of patients are diagnosed and treated.[1] To support the clinical translation of this tracer, Chemistry, Manufacturing and Controls (CMC) must be brought to the current GMP standards of radiopharmacy. The manual radiosynthesis of [^{18}F]Radiocaine has been reported [2] but is not amenable to routine, multi-center clinical production. The route requires two sequential radiochemical steps, each followed by semi-preparative HPLC purification, with an intermediate reformulation. We report the rapid development of a fully automated, GMP-compatible process.
Methods: The synthetic route (Scheme 1) relies on a two-step reaction. [^{18}F]F$^-$ is trapped on an anion-exchange cartridge followed by elution. Subsequent drying of this solution affords [^{18}F]fluoride in its activated form. 1,2-Ethylene ditosylate (**1**) is added to the activated [^{18}F]fluoride. The fluoride anion displaces one of the tosyl leaving group in a SN_2 reaction. This labeling reaction leads to the formation of the crude 2-[^{18}F]fluoroethyl tosylate ([^{18}F]FETs, **2**) which is purified on a semi-preparative column. Following a reformulation to remove the HPLC solvents, the purified **2** is then reacted with (*N*-(2,6-dimethylphenyl)-2-(ethylamino)acetamide) (**3**). The second crude reaction mixture is then also purified on a semi-preparative column, affording [^{18}F]Radiocaine. The radiotracer is formulated, sterile-filtered and ready for quality control. The full sequence was run on a single 24-valve cassette (AllinOne, Trasis).
Results: The automated radiosynthesis and purification of 2-[^{18}F]fluoroethyl tosylate (**2**) was adapted from an existing fixed-tubing method,[3] providing radiochemical yields of 47-52% n.d.c. (n=2). The alkylation reaction proved more challenging, as the conditions described in the original paper yielded only trace amounts of [^{18}F]Radiocaine. As such, a high throughput screening (HTS) approach was used to rapidly assess various solvents and additives to promote the reaction. The chemistry was then further optimized on the radiosynthesizer, screening various conditions of time, temperature, and quantity of reagents. Following implementation and optimization of the second HPLC purification, the optimized process gave [^{18}F]Radiocaine in 9.6 ± 0.8% RCY (n.d.c., n=2) and with a radiochemical purity of >99% at low activity (300-600 MBq). Three validation batches were then performed at high activity (110-260 GBq of starting activity) and gave [^{18}F]Radiocaine with a RCY of 5.5-8.4% (n.d.c.). Most importantly, the products passed all QC tests and remained stable in their formulation over 6 hours (RCP = 98.4-99%).
Conclusions: Our automated process involves a complex sequence of fluoride activation, followed by two synthetic steps and HPLC purifications, and a final formulation. Yet, it is fully automated and provides [^{18}F]Radiocaine in yields sufficient to support clinical-scale production and repeated batch supply. The product profile is compliant with the European (monograph 0125) and US (USP <825>) Pharmacopeias. This new GMP-compliant manufacturing process enables the production of multi-patient doses of [^{18}F]Radiocaine and is specifically designed for use in hospitals and production centers requiring regular manufacturing of this radiotracer.



Scheme 1: Radiosynthesis of [^{18}F]Radiocaine

Development of Novel Dual Function Reagents for CO_2 Transfer Reactions

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Scheme 1: Carboxylation of heteroarenes **1** by dual-function reagents **2**.

Transfer reactions involving CO_2 are an attractive approach for preparing carboxylic acids and isotope labelling. It enables direct transfer of CO_2 from a surrogate to a target molecule, avoiding the use of highly pyrophoric reagents, expensive transition metals and excessive pressures of CO_2 gas.[1] We have previously developed carboxylation strategies for the functionalisation of C-H, N-H and C-X bonds with carboxylate **2** (Scheme 1A).[2,3,4] However, reaction scope was often limited by the basicity of organometallic intermediate **4** (Ph_3CH = 30.6 in DMSO)[5] therefore, substrates with pKa >30.6 required more forcing conditions or were completely unreactive. To facilitate carboxylation of less acidic C-H bonds, a more reactive organometallic intermediate is required. This work aims to develop novel CO_2 transfer reagents that can carboxylate a wide range of substrates.

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Standard Reporting of the Uncertainty in Reaction Yield Applied to a CO_2 Transfer Reaction

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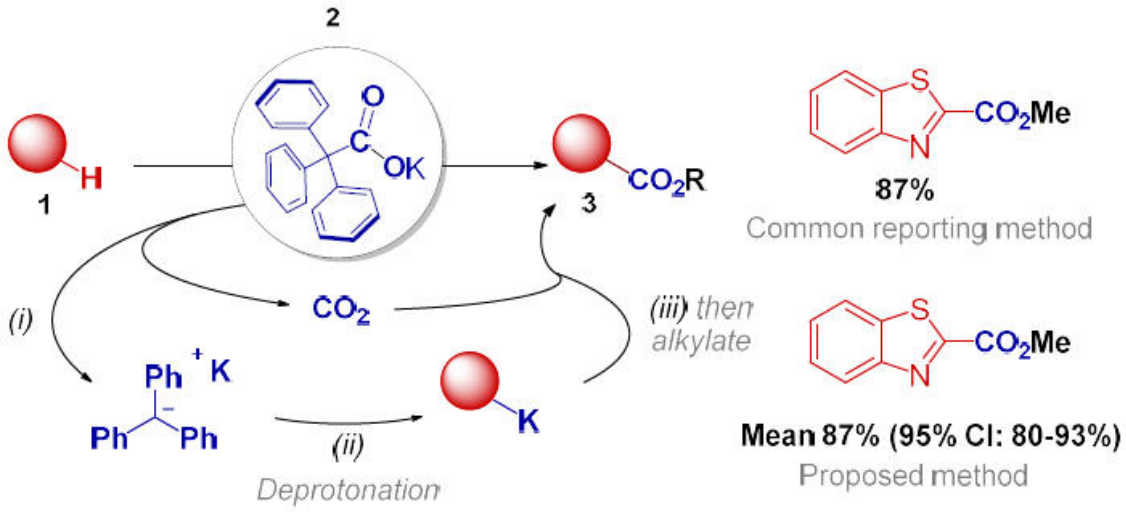


Figure 1: Overview of the carboxylation of different substrates via dual function reagent **2**.

Reproducibility is a central requirement in organic synthesis, yet reaction outcomes are frequently reported as a single value with limited explanation on experimental variability, that would enable others to judge the reproducibility of the reported procedure. Herein we demonstrate a simple, standardised approach for reporting reaction yield as its mean value, accompanied by a confidence interval calculated from repeat experiments.[1] Using several newly developed CO_2 transfer carboxylation reactions as examples, we show how variability plots and confidence intervals provide interpretable measures of yield reproducibility and help identify potential outliers.

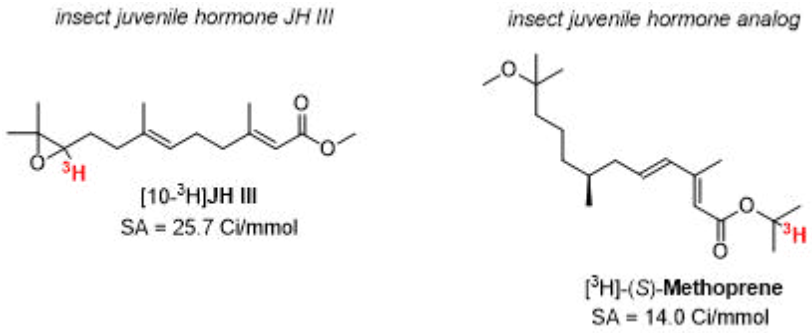
Our previous work on CO_2 transfer carboxylation has demonstrated the potential of dual-function reagents to enable carboxylation under milder conditions, whilst avoiding the direct handling of CO_2 gas.[2] Potassium triphenyl acetate acts as a combined source of base and CO_2, relying on the decarboxylation of the carboxylate salt to afford the trityl anion and CO_2 in situ (Figure 1). Mechanistically, this reactivity suggests a broader potential for carboxylate-mediated CO_2 transfer processes, and we sought to investigate the scope of this transformation. In doing so, we saw this as an opportunity to implement our approach for evaluating the reproducibility of reactions through basic statistical analysis of multiple experimental runs. We make recommendations for best practice in reporting the variability of experimental outcomes and introduce a secure browser-based tool, the Physical Sciences Data Infrastructure (PSDI) Variability Plot Generator, to facilitate transparent reporting of reproducibility.

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Isotopical Labelling of Unfavored Substrates; A Juvenile Hormone JH-111 and its Analog Methoprene

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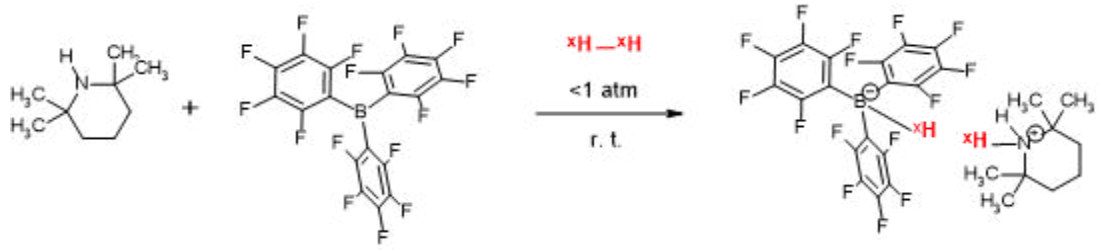
Juvenile hormone (JH) controls insect reproduction and development through an intracellular receptor complex comprising two bHLH-PAS proteins, the JH-binding Methoprene-tolerant (Met) and its partner Taiman (Tai). We examined JH receptors in two representatives of Blattodea, the cockroach *Blattella germanica* and the termite *Prorhinotermes simplex*. To test the JH-binding capacity of Met proteins from these species, we performed chemical synthesis and tritium labeling of the natural juvenile hormone of Blattodea, JH III. An efficient *de novo* radiochemical synthesis of principal natural juvenile hormone JH III used as an important tool to study JH signaling will be shown. Our improved protocol increased the yield and specific activity of [10-^{3}H]JH III (25.7 Ci/mmol) relative to formerly available preparations. Methoprene functions by mimicking natural juvenile hormones necessary for insect development interferes with an insect's life cycle and prevents it from reaching maturity or reproducing. Methoprene is massively applied in Latin America as a larvicide in mosquito control programs to curb the spread of vector-borne diseases such as dengue fever, malaria – dispersed to water bodies, including drinking water reservoirs. This growth regulator has so far been considered non-toxic to mammals. For further studies of its activity in specific ligand-receptor binding studies the hormone was radiolabeled with high specific activity (SA). We will show a [^{3}H]Methoprene synthesis leading to a sufficiently high SA of 14.0 Ci/mmol.
We acknowledge funding from the Czech Academy of Sciences (RVO 61388963).

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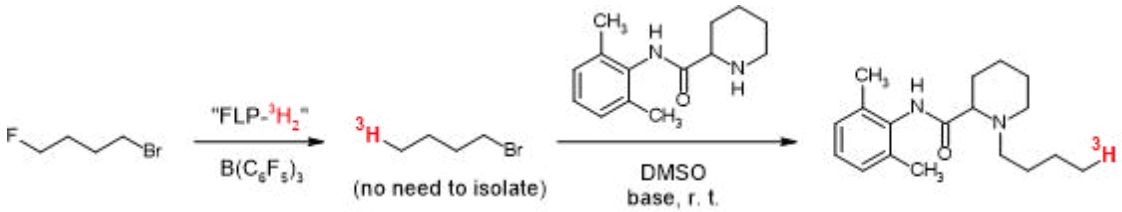
Synthesis of [^{3}H]-Bupivacaine Using FLP-Assisted Tritio-Defluorination

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Scheme 1: A FLP-mediated heterolytic cleavage of ^{x}H–^{x}H bond, forming the FLP-^{x}H$_2$ complex

Borane/amine-based frustrated Lewis pairs (FLPs) display a unique ability to activate molecular hydrogen ^{1}H$_2$, as well as deuterium ^{2}H$_2$ and tritium ^{3}H$_2$, via heterolytic cleavage of ^{x}H—^{x}H bond (Scheme 1), enabling subsequent hydrogen-isotope labeling transformations.
In combination with the high fluorophilicity of the borane component, this reactivity allows highly site-specific replacement of fluorine atom in aliphatic –CH$_2$F group with the corresponding hydrogen isotope.[1] Importantly, other halogen substituents (Cl, Br, I) remain unaffected under these conditions. This unusual chemoselectivity was therefore exploited for the preparation of isotope-labeled alkylating reagents.
The overall transformation was further implemented as a one-pot process, avoiding the need to isolate the tritiated intermediate. The synthetic utility of this approach was demonstrated by the preparation of the local anesthetic [^{3}H]-bupivacaine (Scheme 2).[2] Details, advantages, and challenges of this method will be discussed.



Scheme 2: A pathway leading to tritiated Bupivacaine (0.85 T/molecule, S.A. = 3.14 GBq/mg)

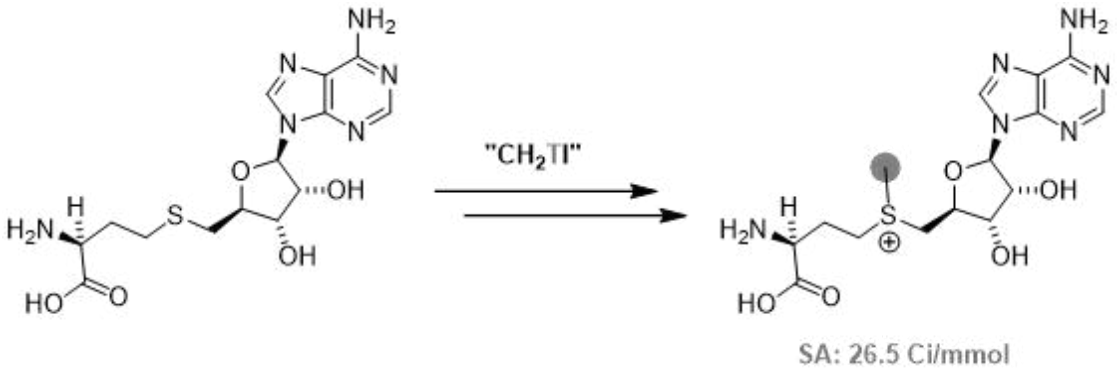
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Preparation of [^{3}H]-*S*-Adenosyl-L-Methionine

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S-Adenosyl-L-methionine (SAM) is a sulfonium-containing primary metabolite found in both prokaryotic and eukaryotic cells. In principle, it is biological methyl-group donor, participating in methylation reactions involving DNA, RNA, proteins, and other biomolecules. SAM is also among the most frequently encountered substrates in enzymatic transformations.[1] Consequently, efficient and reliable methods for the preparation of [methyl-^{3}H]-SAM are of interest in chemical and biochemical research.
Although tritiated SAM is commercially available, its delivery can be unreliable and may take more than six months. One of the synthetic approaches for the preparation of [^{3}H]-SAM involves the methylation of *S*-adenosylhomocysteine with [^{3}H]-methyl iodide. However, [^{3}H]-methyl iodide is very expensive (0.31 nmol, 25 mCi ≈ 4 k$), volatile and unstable reagent. Our long-term research has therefore focused on the preparation and optimization of tritiated methylating agents.[2,3] The key step is reaction of the corresponding [^{3}H]-thioanisole with substituted benzyl iodide at high temperature. Resulting sulfonium salt is decomposed to the desired methyl iodide. We used this efficient and simplified approach to further transformation into a very stable methylating reagent [^{3}H]-methyl tosylate[4] (0.91 T per molecule). The three-step synthesis, the optimization of procedure and the application of other methylating reagents will be discussed.

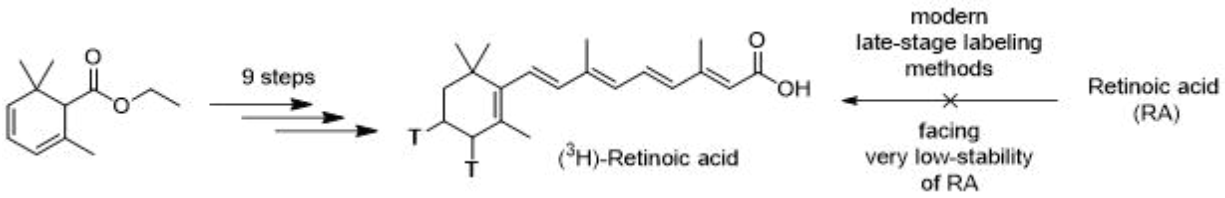
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Challenging Substrate to be Isotopically Labeled; Retinoic Acid

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Scheme 1: Preparation of isotopically labeled all-*E*-retinoic acid

For many years, we have investigated the chemistry of neuropeptides from the insulin/insulin-like growth factor (IGF) signaling system, which are crucial for controlling development, growth, and overall metabolic homeostasis.[1] Expanding upon this work, we aim to investigate retinoic acid (RA), a metabolite of vitamin A, that serves as an essential component of cell-cell signaling during vertebrate organogenesis. RA can also act as a growth factor mimetic capable of replicating, amplifying, or bypassing pathways typically triggered by classical growth factors. Unlike these proteins, which bind to cell-surface receptors, the lipophilic RA diffuses through the cell membrane directly to nuclear receptors, thereby immediately influencing gene transcription. This cross-talk between RA and growth factor signaling allows it to mimic or modulate several key biological processes.

Isotopic labeling is a key tool for the detailed characterization of RA binding properties in various assays. However, retinoic acid is intrinsically unstable and readily degrades when exposed to light, heat, oxygen, and heavy transition metals due to its conjugated polyene chain.[2] Herein, we discuss the hydrogen isotope labeling of this challenging substrate using both modern late-stage labeling methods and a synthetic pathway yielding
a suitable precursor for subsequent selective tritiation.

We report a nine-step synthesis of all-*E*-retinoic acid starting from commercially available ethyl 2,6,6-trimethyl-2,4-cyclohexadiene-1-carboxylate. The labeling was achieved in the penultimate step via selective hydrogenation using a noble metal catalyst. This highly selective approach enables the preparation of [^{3}H]-RA, yielding the most stable tritiated form of retinoic acid reported to date.

This work was supported by Czech Academy of Sciences (RVO 61388963).

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Haidian District
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Contact: Yuan Congying
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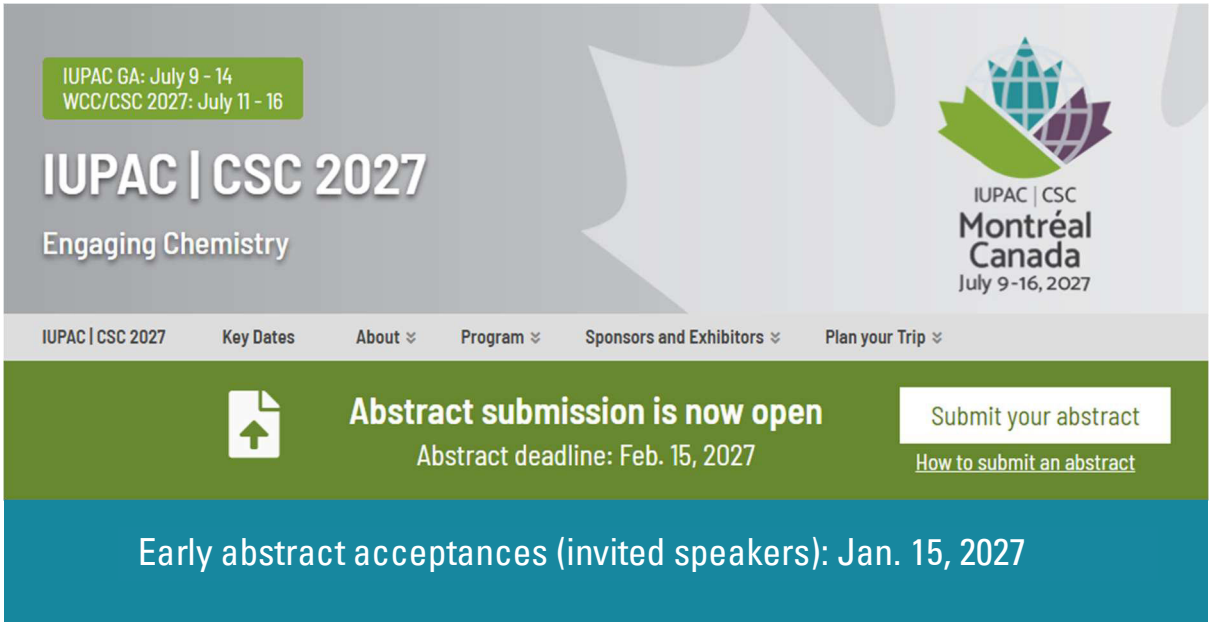
Contact: Andrew Ditchman
Tel: 01371 831611
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List of Delegates

The following attendees have agreed for their details to be included in the conference booklet.
Delegates not wishing to share their information have not been included.

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Symposium: **Emerging Green and Sustainable Strategies for Isotope Labelling**
IUPAC Theme: **Molecular Scale** (*for submissions*)

Academic and industrial researchers are invited to contribute oral and poster presentations to this exciting symposium which will explore **emerging sustainable, energy and resource-efficient approaches to isotope labelling** i.e., to showcase innovative approaches to making isotope labelling of compounds more environmentally sustainable, energy- and resource-efficient, and scalable.

The programme will bring together researchers working across stable and radioactive isotope labelling, including deuterium, tritium, carbon-13, carbon-14, fluorine-18, nitrogen-15, oxygen-18, and other relevant isotopes. Topics will span green synthetic methodologies such as biocatalysis, mechanochemistry, electrochemistry, photochemistry, catalyst-free, continuous-flow processing, automation and robotics, and AI/ML-enabled reaction design and optimisation. The symposium also welcomes talks which explore sustainability metrics and life-cycle approaches tailored to isotope chemistry, as well as emerging approaches to recover and reuse catalysts and isotope-labelled reagents.

For further information, please contact **Prof Wim de Borggraeve, KU Leuven** (wim.deborggraeve@kuleuven.be) or **dr. Ross Jansen-van Vuuren, University of Ljubljana** (Ross.JansenvanVuuren@fkkt.uni-lj.si).

Hope to see you in Montreal in 2027!

ALMAC

^{14}C Radiolabelling

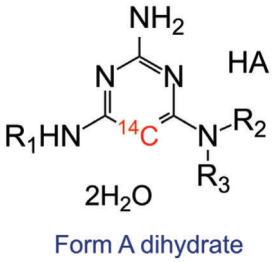
Solving complex radiolabelling challenges

Serious about difficult science



Case Studies

Case Study 1 – Maintaining Physical Form During GMP Radiolabelling



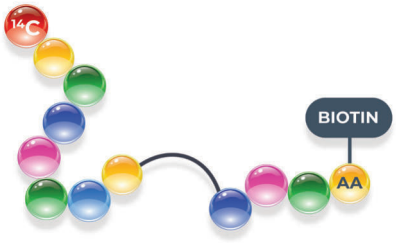
Form A dihydrate

Challenge
A client required 3 mCi of ^14C-labelled drug substance manufactured under GMP conditions while maintaining a specific polymorphic form throughout manufacture and release.

The Science
Our team integrated radiolabelling, GMP manufacturing and physical sciences expertise to develop a controlled process incorporating crystallisation, drying, milling and hydration steps. Physical form was monitored and confirmed throughout manufacture using XRPD analysis.

Outcome
The desired polymorphic form was successfully isolated and confirmed. GMP-labelled material was delivered together with release testing and stability data to support the client's programme.

Case Study 2 – Preserving Valuable Labelled Amino Acid in Peptide Synthesis



Challenge
A client required 2 mg of a ^14C-labelled peptide while minimising consumption of an expensive labelled amino acid.

The Science
Radiolabelling and peptide synthesis specialists worked together to redesign the critical coupling strategy. The synthetic route was optimised to maximise incorporation efficiency while reducing losses of the labelled building block.

Outcome
The peptide was successfully synthesised while preserving valuable labelled material, helping minimise cost and ensuring successful project delivery.

Case Study 3 – Integrating Multiple Disciplines for PDC Manufacture

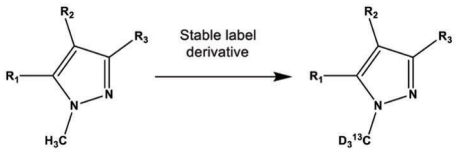


Challenge
A programme required radiolabelling of linker technology followed by manufacture of a peptide-drug conjugate, creating significant complexity across multiple scientific disciplines.

The Science
Experts from radiolabelling, biology and purification collaborated to develop an integrated manufacturing solution. Advanced purification strategies incorporating preparative HPLC, HIC chromatography and ultrafiltration were used to achieve the required product quality.

Outcome
The client received purified labelled PDC material that met programme requirements, demonstrating the value of a fully integrated scientific team.

Case Study 4 – Achieving High Isotopic Purity for Stable Isotope Labelling



Challenge
client required 250 mg of ^13CD3-labelled material with confidence in isotopic incorporation and impurity profile.

The Science
The synthetic route was developed to produce the labelled compound efficiently, followed by preparative HPLC purification and comprehensive mass spectrometry analysis to evaluate isotopic purity and quantify unlabelled species.

Outcome
High-quality labelled material was delivered together with detailed isotopic purity characterisation, enabling confident progression of the client's programme.

sciences@almacgroup.com

ARCINOVA
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Isotope labelling

40+

years of excellence in isotope labelling

98%

on average delivery success rate

500+

molecules synthesised

- **Supporting non-clinical and clinical studies.**
- **Available as a stand-alone service or as part of a holistic ADME programme delivered with Quotient Sciences.**

A proven track record of ensuring the highest precision and quality in radiolabelled compounds

Accelerate your non-clinical and clinical ADME studies with high-purity, fit-for-purpose ^{14}C and stable isotope-labelled compounds, expertly designed to support metabolism studies and quantification in biological matrices. With over 40 years of on-site radiochemistry experience, Arcinova, A Quotient Sciences Company offer trusted consultancy and end-to-end delivery, backed by a 98% on-time success rate.

Our specialist chemists apply deep expertise in radiochemical, analytical, and synthetic chemistry to guide optimal label placement across a broad range of molecular entities.

Our approach is built for faster timelines, fewer handoffs, and reliable results that keep you on-track towards your development goals.



Synthesis-to-Clinic®: Integrated ^{14}C-labelled drug substances and drug products with full ADME programme delivery

Our ^{14}C-labelling capabilities are available as a standalone service through Arcinova or as part of an integrated human ADME programme delivered by the **Quotient Sciences Synthesis-to-Clinic® platform**—combining all steps needed to conduct a full ADME clinical programme from radiolabelling and clinical study through to the final clinical report.

Expert-led labelling strategy

Optimal ^{14}C label positioning in a metabolically stable part of the molecule is crucial to ensure key metabolites are identifiable to provide adequate ADME data. We offer upfront consultancy on labelling strategy, then design a synthetic strategy that delivers high isotopic purity and meets the requirements of your study.

Additionally, we support projects by synthesising non-clinical and clinical batches separately or by producing a preclinical intermediate for both. This integrated approach can eliminate re-synthesis, saving time and reducing costs for human ADME studies.

We are experts in the safe handling of highly potent, hazardous, and cytotoxic compounds and are licenced for Scheduled I-IV controlled substances, including cannabinoids, ketamine, and tryptamine.

Our experience also includes:

- Synthesis of stable-labelled compounds incorporating ^{2}H, ^{13}C, ^{15}N or ^{18}O, ensuring optimal label placement and high isotopic purity
- Expertise in polymorphic control, particle size reduction, lyophilisation, and polymer dispersion for poorly soluble radiolabelled drugs
- Synthesis of metabolites and reference materials tailored for bioanalytical and metabolism research

Specialist radiolabelling, delivered end-to-end

- MHRA-accredited, offering full GMP synthesis of ^{14}C-labelled compounds for human studies
- Proven expertise in handling high-potency compounds, controlled substances, and volatile ^{14}C gases ($^{14}CO_2$, ^{14}CO, $^{14}COCl_2$)
- Dedicated team including a technical chemist lead, analytical lead, and project manager to ensure consistent delivery
- Fully integrated support through our all-in-one facility – spanning isotopic labelling, drug substance development, formulation, IMP manufacture, and dossier preparation
- Microbiological and solid-state analysis (PSD and XRPD) to support all stages of development

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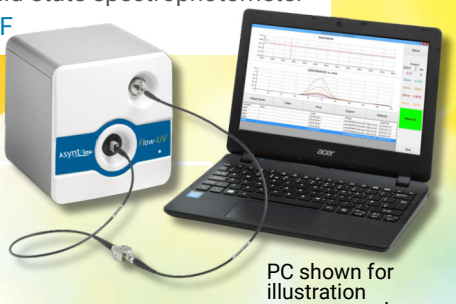
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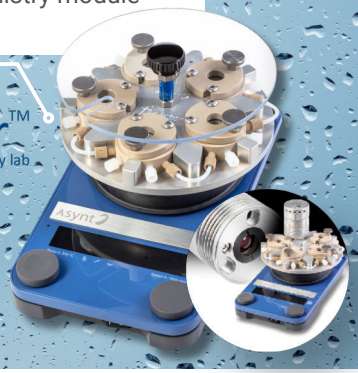




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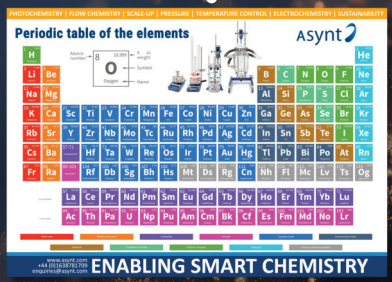
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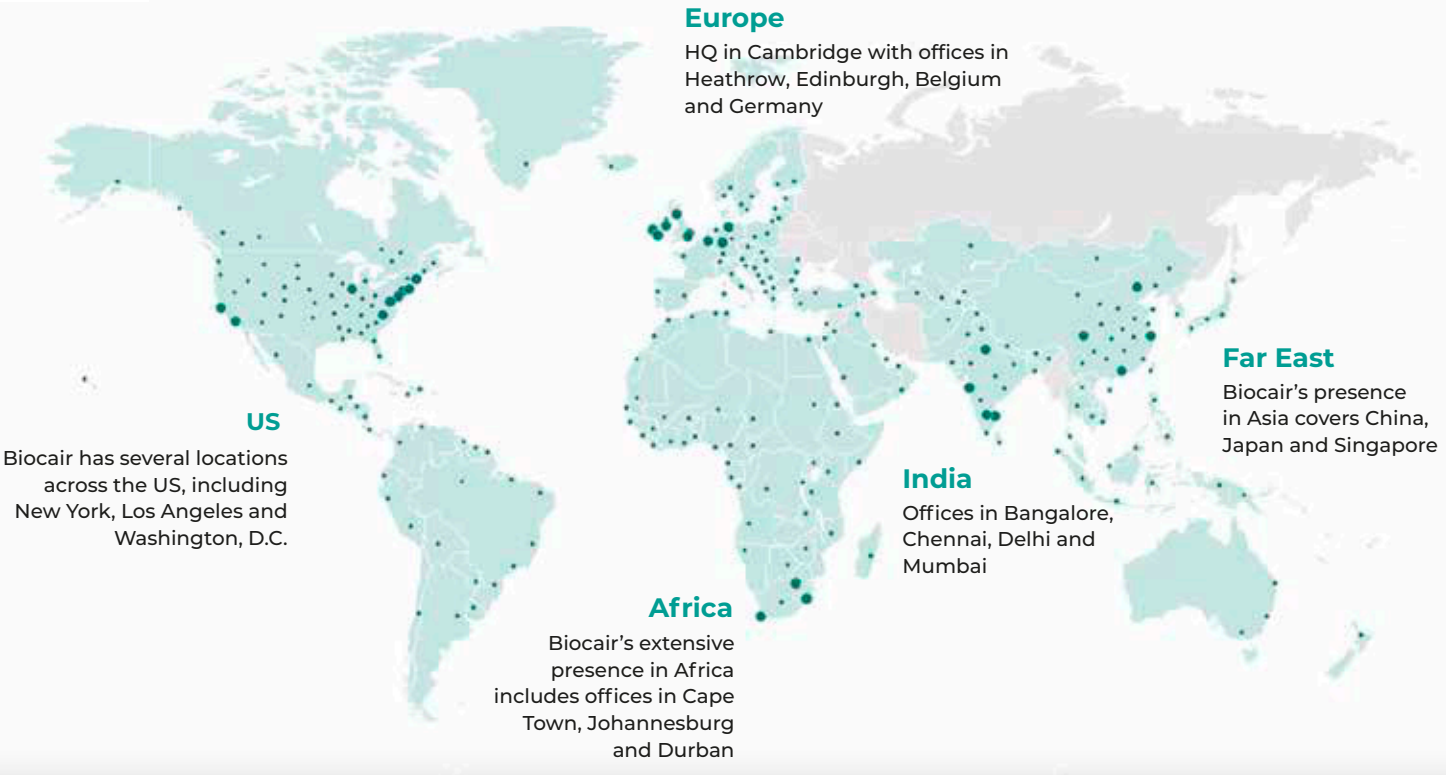
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C-14 Barium Carbonate

1. Product describtion:

Product Name: C-14 Barium Carbonate ($Ba^{14}CO_3$)

Radioisotope: Carbon-14 (^{14}C)

Form: Dry powder

2. Product specification:

Specific Activity, mCi/g : ≥250

Radionuclide purity, % : ＞99%

Radiochemical purity, % : ＞99%

Chemical purity, % : ≥95%

3. Storage condition: at an ambient temperature

4. Country of origin: China

5. Contact information: yuancongying@circ.com.cn



Carbon-14 Custom Synthesis Service

1. Small Molecule Compounds

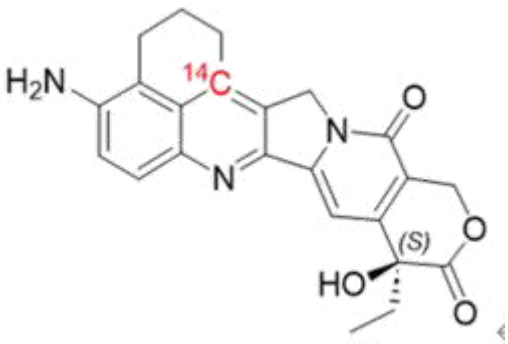
- **Benzene ring-containing compounds**

  Isotopic labeling is available with either a single carbon-14 atom attached to the benzene ring, or uniform labeling (Ph-U).

- **Pyridine ring-containing compounds**
- **Pharmaceutical chemicals**

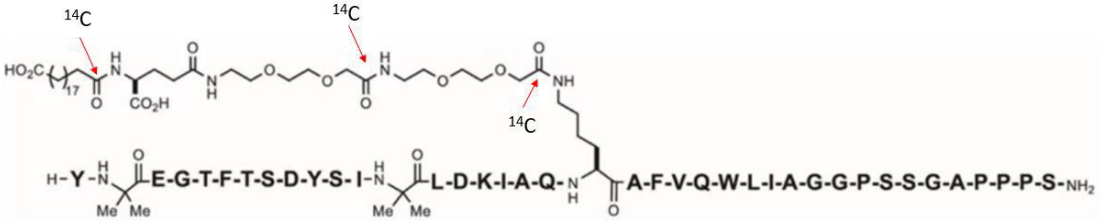
2. Payload-Dxd Analogues

(The diagram below is a schematic representation, not an actual case)



3. Peptides

(The diagram below is a schematic representation, not an actual case)



4. Contact information: yuancongying@circ.com.cn

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- Carbon-14 radioisotope labelling for Life Sciences
- Carbon-14 GMP API synthesis for human AME studies
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- Stable isotope, metabolite & process impurity synthesis and fermentation
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Standard Cell Model

For users that require a traditional flow cell configuration, the Standard Cell (SC) provides the option of using either liquid mixing or solid scintillator cells.

The standard cell model can be operated horizontally, as a stand-alone unit, or as part of an HPLC stack. It can also be placed in a space-saving vertical position alongside the HPLC.

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  - Optimization of the radiosynthetic pathway and yield
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  - GMP or GLP certification available

Clinical hAME studies with ^{14}C Compounds

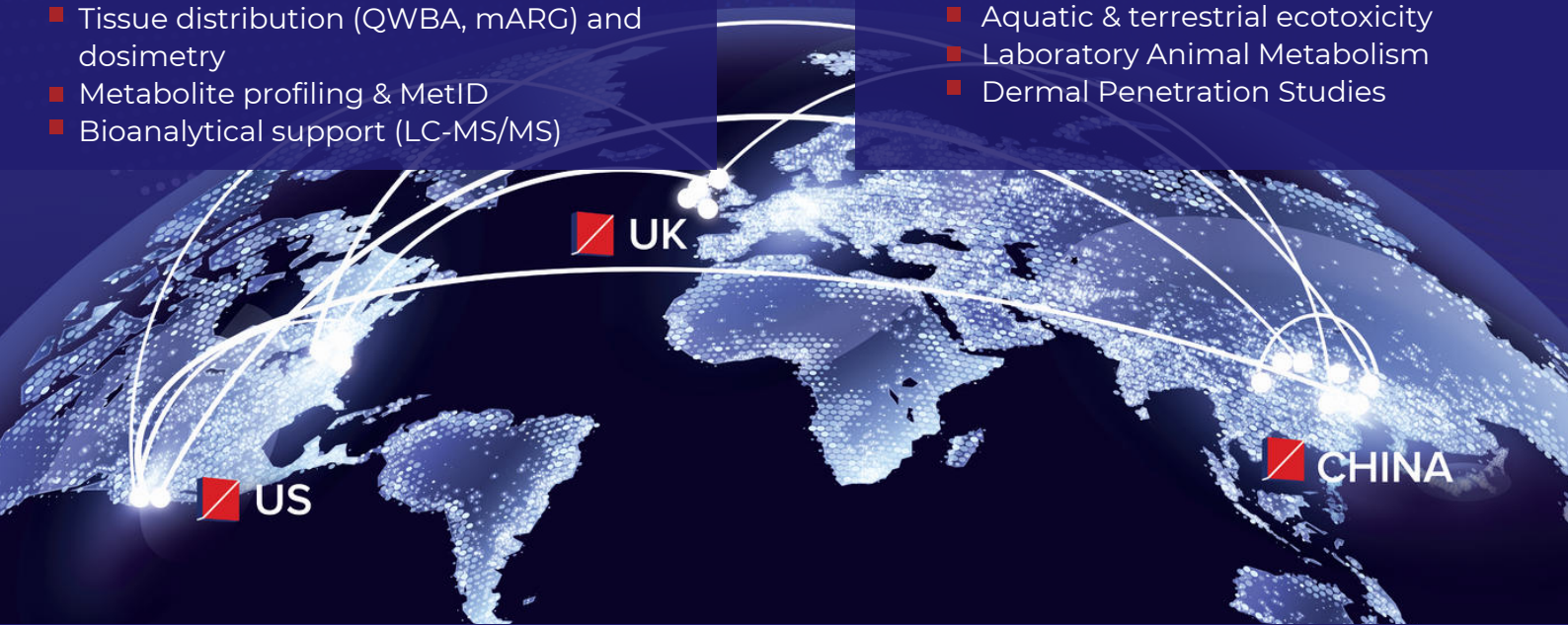
- Absorption, metabolism and excretion for mass balance & PK (AME)
- IV microtracers for absolute bioavailability (ABA)
- AMS (Accelerator Mass Spectrometry)
- High resolution mass spectrometry (HRMS) for MetID
- Bioanalysis and biomarkers (LC-MS/MS, immunoassay)

Non-clinical ADME & QWBA with Radiolabelled Compounds

- Pharmacokinetics, biliary excretion & excretion balance
- Tissue distribution (QWBA, mARG) and dosimetry
- Metabolite profiling & MetID
- Bioanalytical support (LC-MS/MS)

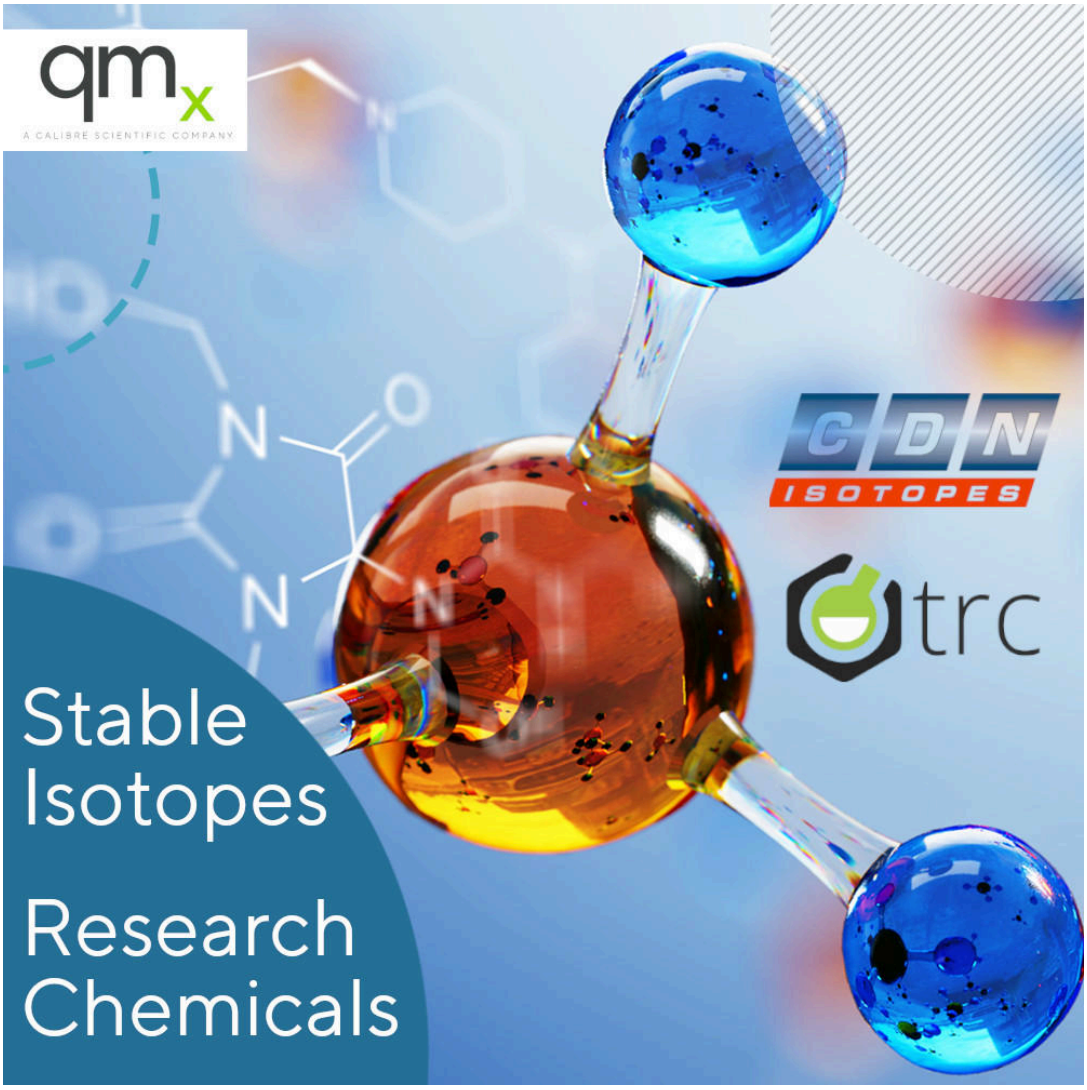
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